Research Statement

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Research overview

My research primarily focuses on topological data analysis (TDA), which uses ideas from algebraic topology to identify qualitative features such as connected components, loops, and higher-dimensional structures in complex data. A central tool in TDA is persistent homology (PH) (see Fig. 1 and the accompanying interactive tool[1]), which records when these features appear and disappear as a scale parameter varies, encoding this information in a *persistence diagram* or a *persistence barcode*. I study TDA from both theoretical and applied perspectives: my dissertation explores the metric geometry of persistence diagram spaces, while my applied work focuses on applications of TDA to neuroscience, using topological features to decode neural population activity.

Two publications form the core of my dissertation. First is a joint work with my advisor Tom Needham where I study the metric geometry of persistence diagrams and prove that the two important distances in TDA (namely, the erosion and landscape distances) are equal [2]. This result connects the categorical interpretation of the erosion distance with the computational and statistical accessibility of persistence landscapes, allowing insights and techniques developed in one framework to be transferred to the other.

For my second publication, on which I am the co-first author, I developed a PH pipeline for analyzing simultaneously recorded neural spike trains in collaboration with an interdisciplinary team of researchers in mathematics and neuroscience [1]. Our experiments show that the topological pipeline is particularly effective when information is encoded primarily in coordinated activity across neurons rather than in individual firing rates.

These projects have given me valuable experience in tackling problems and provide opportunities to extend this line of work. Looking forward, I am interested in connecting TDA with machine learning and deep learning to study multivariate time series, particularly those arising in biological problems. Below, I describe my completed work, ongoing collaborations, and planned research directions.

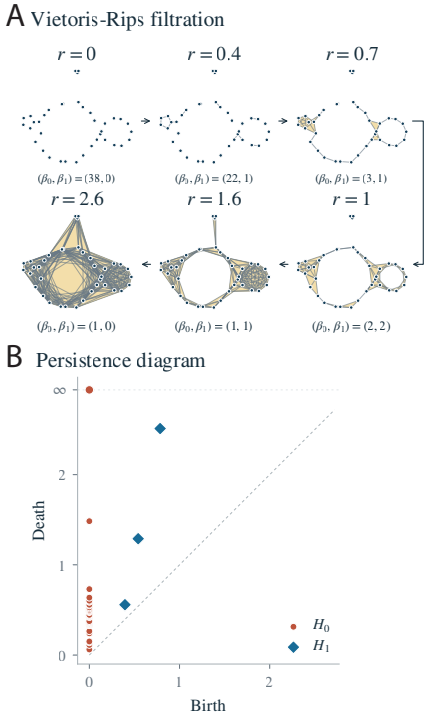


Figure 1: **Persistent homology of a Vietoris–Rips filtration. (A)** Vietoris–Rips complexes $\mathrm{VR}(X; r)$ built on a point cloud (X, d) at increasing scales r. Edges connect points within distance r, and triangles fill when all three edges are present. Connected components merge while loops form and fill. Below each complex, (β_0, β_1) gives the number of connected components and independent loops, respectively. **(B)** The corresponding persistence diagram records the birth and death scales of these features.

[1] https://cagatay-ayhan.github.io/Visual-TDA/docs/PH_demos/persistent_homology_animation.html

1. Metric geometry of persistence diagrams

Persistent homology summarizes a dataset across scales by a persistence diagram, a finite multiset $Y = \{(b_i, d_i)\}_{i=1}^{N}$ recording the birth and death times of topological features [27, 8] (see Figure 1). Statistical and machine learning applications require meaningful ways to compare these diagrams. Consequently, there has been considerable recent interest in studying different metrics on the space of persistence diagrams. Understanding these metrics and their relationships helps transfer theoretical guarantees and computational tools between them.

Three metrics motivate my work. First, the *bottleneck distance*, d_B [14, 11], compares persistence diagrams by matching their points minimizing the largest L^∞ displacement. It is a widely popular metric within the TDA community due in part to its natural interpretation in the language of category theory and its stability (small perturbations of the data produce small changes in the diagrams). Secondly, Bubenik's *landscape distance* [5], d_L, arises from an embedding of persistence diagrams into a Banach space. This embedding, known as the *landscape map*, assigns to each persistence diagram Y, a *persistence landscape*, $\lambda^Y = (\lambda_1^Y, \lambda_2^Y, \dots)$, which is a sequence of piece-wise linear functions. Each birth-death pair $(b_i, d_i) \in Y$ defines a triangular 'tent' function supported on $[b_i, d_i]$; at each $t \in \mathbb{R}$, $\lambda_k^Y(t)$ takes the kth-largest value among these tents, or zero if fewer than k tents are present (see Fig.2B). The distance d_L is the L^∞-norm of the difference between the resulting landscapes. This functional representation also allows diagrams to be studied using standard statistical methods e.g. their landscapes can be summed and averaged. Lastly, Patel's *erosion distance*, d_E, compares persistence diagrams through their rank functions: the rank of a persistence diagram Y at (b, d), with $b < d$, counts the features that persist throughout $[b, d]$, namely the birth–death pairs $(b_i, d_i) \in Y$ satisfying $b_i \le b < d < d_i$. An ε-erosion of a persistence diagram moves each birth coordinate ε units to the right and each death coordinate ε units down, replacing (b_i, d_i) with $(b_i + \varepsilon, d_i - \varepsilon)$. Features whose death is then at or before their birth are discarded. The distance d_E is the infimum over all $\varepsilon \ge 0$ such that each eroded diagram's rank function is pointwise no larger than the other diagram's original rank function.

Known relationships between these three metrics help us study their properties. For example, the inequalities $d_E \le d_B$ [22] and $d_L \le d_B$ [5], imply that erosion and landscape distances inherit the stability guarantees of bottleneck distance. Moreover, these inequalities can be strict, motivating a closer examination of when the metrics coincide and how their geometric properties are related. In fact, these questions are central to my dissertation and have led to new connections between the erosion, landscape, and bottleneck distances. I highlight several of these results below.

In joint work with Needham [2], we prove that the erosion and landscape distances are exactly the same:

Theorem A (Equivalence of Erosion and Landscape Distances)**.** The landscape distance between any two persistence landscapes is equal to the erosion distance between their corresponding persistence diagrams. That is, $d_L(Y, Y') = d_E(Y, Y')$ for any pair of persistence diagrams $Y, Y' \in \mathsf{PDgm}$.

To prove Theorem A, we adopt the *category with a flow* framework [13] and its counterpart, a *coflow* [12]. Put simply, equipping a category with a coflow (or flow) gives rise to an extended pseudometric called the interleaving distance. Thus, we define poset categories of persistence diagrams and persistence landscapes, equipping them with coflows whose interleaving distances recover the erosion and landscape distances, respectively. Along the way, this approach reveals new structural properties of persistence landscapes. Building on the work of Betthauser, Bubenik, and Edwards [4], we give the first characterization of the image of the landscape map stated entirely independently of persistence diagrams (see [2, Def. 3.11, Cor. 3.14]). To complete the proof, we show that the landscape map is an order isomorphism that preserves the coflows. This implies that the induced interleaving distances coincide, so the erosion and landscape distances are equal.

We also study whether persistence diagrams equipped with these metrics are length spaces or geodesic spaces. These notions come from metric geometry (see [7]), and we briefly recall them here. The length of a continuous path in a metric space (X, d) is defined by summing the distances between successive points sampled along the path and taking the supremum over all partitions of its parameter interval. The corresponding *intrinsic distance*, $\hat{d}$, between two points is the infimum of the lengths of all continuous paths joining them. If $\hat{d} = d$, then (X, d) is called a *length space*. If every pair of points can be joined by a path whose length equals their distance, then (X, d) is called a *geodesic space*.

The space of persistence diagrams equipped with the bottleneck distance is known to be geodesic [10, 9]. In the following theorem, we show that the intrinsic distance associated with the erosion (landscape) distance is the bottleneck distance. Since $d_E = d_L \neq d_B$, the erosion (landscape) distance does not define a length space and therefore does not define a geodesic space. This answers a question posed by Kim and Mémoli [15, Example 3.15]:

Theorem B. The intrinsic metric corresponding to erosion (landscape) distance $\hat{d}_E = \hat{d}_L$ is bottleneck distance d_B. That is,

$$\hat{d}_E(Y, Y') = \hat{d}_L(Y, Y') = d_B(Y, Y')$$

for all $Y, Y' \in \mathsf{PDgm}$.

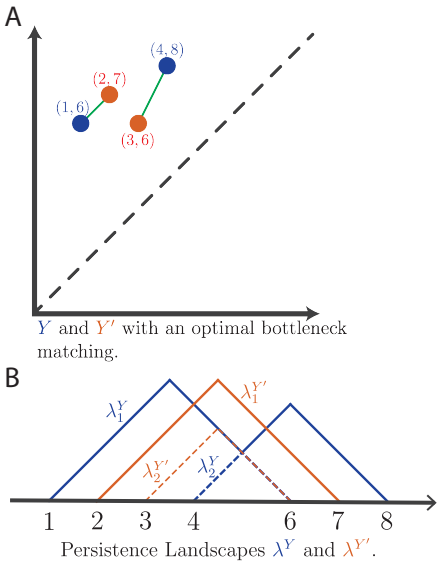


Figure 2: Example persistence diagrams Y and Y' illustrating $d_E(Y, Y') = d_L(Y, Y') = 1 < 2 = d_B(Y, Y')$. **(A)** Diagrams $Y = \{(1, 6), (4, 8)\}$ (blue) and $Y' = \{(2, 7), (3, 6)\}$ (orange). Green segments show an optimal matching, giving $d_B(Y, Y') = 2$. **(B)** Corresponding persistence landscapes (first functions solid, second dashed). The maximum vertical difference between functions is 1 (e.g., at $t = 2$ or $t = 4$), giving $d_L(Y, Y') = 1$. The erosion distance $d_E(Y, Y')$ yields the same value.

Finally, we study a specific subspace of persistence diagrams referred to as *birth-zero diagrams*. These are diagrams of the form $\{(0, d_i)\}_{i=1}^N$ where the birth coordinates are fixed to 0, and the death coordinates are sorted in descending order $d_1 \geq d_2 \geq \cdots \geq d_N > 0$. We denote this subspace by PDgm^0. Such diagrams arise naturally as the finite part of degree-zero persistent homology of Vietoris–Rips filtrations. We leverage the convenient structure of birth-zero diagrams to prove the following two results.

Theorem C. For any $Y, Y' \in \mathsf{PDgm}^0$, we have $d_B(Y, Y') = d_E(Y, Y') = d_L(Y, Y')$.

Finally, we use the structure of birth-zero diagrams to study coarse embeddability of the space of persistence diagrams into Hilbert spaces. Roughly, a course embedding is a highly relaxed notion of an isometric embedding, which concers preservation of large-scale features. The question of coarse embeddability of the space of persistence diagrams is important for downstream statistical analysis operations. These questions have attracted substantial attention. Bubenik and Wagner showed that persistence diagrams equipped with the bottleneck distance do not coarsely embed into any Hilbert space [6]. Wagner extended this non-embeddability result to p-Wasserstein distances for $p > 2$ [20]. On the other hand, Mitra and Virk showed that spaces of diagrams with uniformly bounded cardinality do admit coarse embeddings under bottleneck and Wasserstein distances [17]. The 2-Wasserstein case remains open; Pritchard and Weighill showed that it is equivalent to the corresponding question for probability measures on the plane [18]. Our work strengthens the bottleneck non-embeddability result by showing that the obstruction persists even for birth-zero diagrams, and extends it to the erosion and landscape distances.

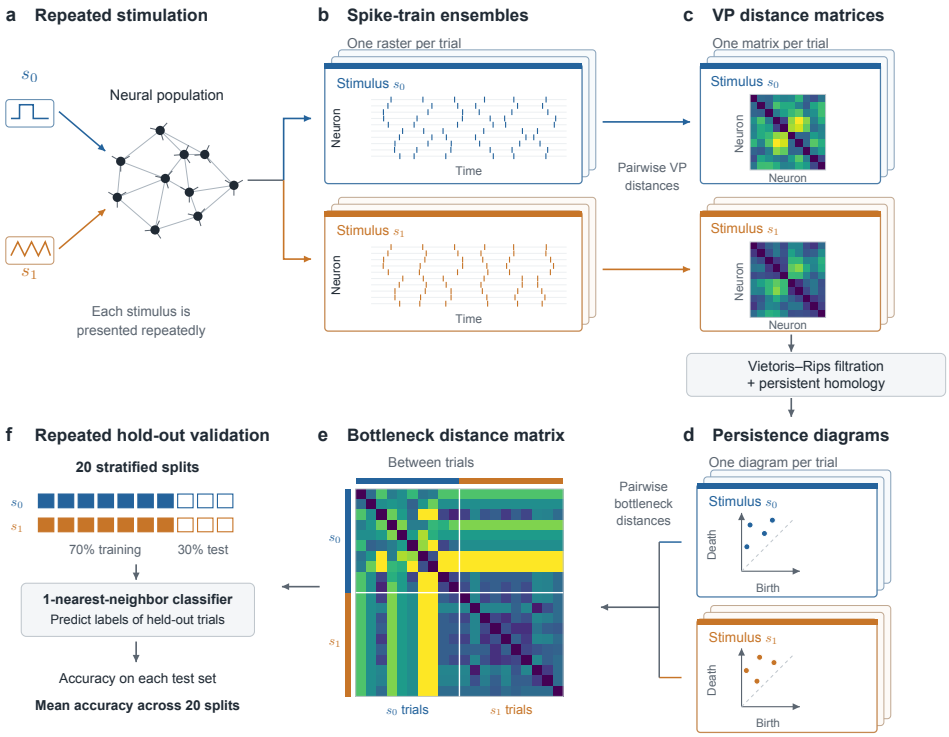


Figure 3: A Persistent Homology Pipeline for the Analysis of Neural Spike Train Data

Theorem D (Coarse Non-Embeddability in Hilbert Spaces)**.** The space $(\mathsf{PDgm}^0, d_\bullet)$ of birth-zero persistence diagrams, equipped with the distance $d_\bullet = d_B = d_E = d_L$, does not admit a coarse embedding into a Hilbert space. Consequently, the full spaces of persistence diagrams $(\mathsf{PDgm}, d_\bullet)$, with $d_\bullet = d_B$ or $d_\bullet = d_E = d_L$, also fail to admit such an embedding.

2. Topology of coordinated neural activity

Thanks to modern electrophysiology, neuroscience produces vast amounts of simultaneous recordings from neuronal ensembles. This allows researchers to study coordinated neural population activity, which is critical to understanding how the brain encodes information. The complexity of these data motivates mathematical approaches such as persistent homology.

In joint first-authored work with researchers in mathematics and neuroscience [1], I developed a persistent homology pipeline, shown in Figure 3, for analyzing simultaneously recorded neuronal ensembles. I wrote and maintain the analysis code for this project.[2] Our experiments show that this TDA approach is particularly well suited to classification when information is encoded primarily in coordinated activity across neurons rather than in the activity of individual neurons. The data we analyze are obtained as follows. A network of neurons is exposed to repeated presentations of different stimuli (Panel a); for example, a mouse receives water at different temperatures while neurons in its gustatory cortex are recorded simultaneously.

During each stimulus presentation, each neuron's activity is recorded as a *spike train*: a set of firing times $\mathsf{S} = \{t_1, \ldots, t_n\}$, where $t_i \in \{0, 1, \ldots, T\}$ and the positive integer T denotes the final recording time step. A *train ensemble* is a collection of spike trains recorded simultaneously during one trial, displayed as rows in Panel b. We treat train ensembles as unordered because our TDA analysis does not require a

[2] https://github.com/cagatay-ayhan/Persistent-Homology-Pipeline-for-Neural-Spike-Train-Data

correspondence between individual neurons in different recordings.

To study the global "shape" of population activity, we compare spike trains using the Victor–Purpura (VP) distance [19]. It is the minimum cost of transforming one spike train into another by adding or deleting spikes at unit cost, or shifting them at a cost of q times the temporal displacement. The parameter $q \geq 0$ controls sensitivity to spike timing. This allows us to view each train ensemble as a finite metric space [3], represented by the matrix of pairwise VP distances between its spike trains (Panel c). We then apply Vietoris–Rips persistent homology to obtain a persistence diagram in a chosen homological degree for each ensemble (Panel d). These diagrams provide topological summaries of the neural activity.

To assess whether these summaries retain information that distinguishes the stimuli, we compute pairwise bottleneck distances between the diagrams (Panel e) and apply a 1-nearest-neighbor classifier. We use a repeated hold-out validation scheme; that is, we evaluate the model using 20 stratified 70:30 train/test splits, preserving stimulus proportions (Panel f). Each test trial receives the stimulus label of its nearest training trial in bottleneck distance. Mean test accuracy across splits measures how well the summaries retain stimulus information.

Theoretical Results

Our pipeline includes two stability results. The first shows that small changes in the VP distance parameter q lead to small changes in the persistence diagrams. This result follows as a corollary, as stated below, from a more general stability theorem we prove for families of pseudometrics.

Corollary E. Let $\mathcal{R}$ be a train ensemble with spike times in $\{0, 1, \ldots, T\}$. For all $q, q' \geq 0$ and homological degrees $m \geq 0$,

$$d_B(\mathcal{B}_m(\mathcal{R}, \mathrm{VP}_q), \mathcal{B}_m(\mathcal{R}, \mathrm{VP}_{q'})) \leq \frac{T(T+1)}{2}|q - q'|,$$

where $\mathcal{B}_m$ denotes the degree-m Vietoris–Rips persistence diagram.

This gives an explicit bound on how much the diagrams can change with q. For our computations, we choose $q > 2$, where the VP distance simplifies to $\mathrm{VP}_q(\mathsf{S}, \mathsf{S}') = |\mathsf{S} \setminus \mathsf{S}'| + |\mathsf{S}' \setminus \mathsf{S}|$ and is independent of q.

The second stability result concerns changes in the probability distributions governing neural responses. Intuitively, if a network responds similarly to two stimuli, the corresponding distributions of persistence diagrams should also be similar. I omit the details here, but the takeaway is that, as the number of independent experimental trials from the same response distribution increases, the empirical distribution of TDA signatures is guaranteed to converge to that of the underlying population. However, convergence may be slow, and an accurate estimate may require a large number of trials.

Results With Synthetic and Biological Data

In the journal version of the paper, to appear in the *SIAM Journal on Life Sciences* (SIALS), we present experiments demonstrating that our TDA pipeline is particularly well suited to detecting coordinated neural population activity. We compare its performance with support vector machines (SVMs) classifier.

In synthetic examples, it distinguishes stimuli encoded through coordination even when single-neuron classifiers and SVM perform near chance. Another example demonstrates the value of detecting loops: degree-1 PH distinguishes cyclic from noncyclic activity when degree-0 PH and SVM do not. Our largest synthetic experiment is one where we simulated a population of 100 neurons responding to three distinct stimuli. A certain portion of the population encodes information by varying their coordination (referred to as *correlation coders*), and another group that encode information by varying their individual firing

[3] The VP distance is a pseudometric when $q = 0$ and a metric when $q > 0$.

rates depending on the stimulus (referred to as *rate coders*). TDA performs better when correlation coders carry the stimulus information, while SVM performs better as information in individual firing rates becomes more prevalent.

Biological recordings provide further evidence. In mouse gustatory cortex [4], TDA exceeds the average single-neuron classification accuracy in 30 of 38 recording sessions, although SVM usually performs better. TDA also remains unchanged when neuron labels are permuted across trials, whereas SVM performance generally decreases. In macaque visual cortex [23, 24].[5], we test coordination more directly by distinguishing actual simultaneously recorded responses from trial-shuffled responses, which preserve individual spike trains while disrupting their coordination and simultaneity. TDA outperforms an SVM with a radial basis function kernel in 38 of 45 session–stimulus pairs.

For future work, my goal is to make the pipeline time-resolved. Currently, we summarize each train ensemble over the full recording window. This summary is not injective: distinct ensembles can produce identical persistence diagrams. For example, ensembles with very different overall firing rates can have identical pairwise VP distances, and therefore identical persistence diagrams. This raises a mathematical question: which differences in neural activity are lost in the topological representation, and what additional information would allow us to distinguish them?

3. Ongoing and Future research

TDA for Materials Science. During my Summer 2026 internship at the Georgia Tech Research Institute, supervised by Branden Stone, I explored applications of TDA to materials science. Preliminary experiments revealed relationships between Betti curves and specific material properties. This ongoing collaboration asks whether these topological summaries can help identify promising dopants (impurities added to materials to change their properties). The project has broadened my experience applying TDA in a new scientific domain.

Generalizing Gromov–Hausdorff Distance. In joint work with my advisor, Tom Needham, I am developing a generalization of Gromov–Hausdorff distance inspired by the Z-Gromov–Wasserstein framework [3]. We consider Z-valued structures, which assign a value in a metric space Z to each finite subset of a set. This framework includes ordinary metric spaces while allowing us to describe relationships among groups of points, beyond pairwise distances. Our goal is to adapt the standard Gromov–Hausdorff distance to the setting of these structures. Potential applications include stability in multiparameter persistent homology and detecting communities in attributed networks.

Topological Deep Learning for Multivariate Time Series. Transformers have attracted significant interest in time series modeling (e.g. [21, 26]), and recent work has explored incorporating topological information into these models (e.g. [25, 16]). In ongoing work with collaborators at Florida State University, I am investigating how topological information can improve transformer-based models for multivariate time series. We aim to identify when topological features capture relationships that these models might otherwise miss and whether incorporating such features improves predictive performance. This research also links to the extension of my work on neural spike train data.

Overall, my long-term goal is to develop geometric and topological tools and explore how they can be integrated with transformers and other deep learning methods to address scientific questions. I am particularly interested in biological applications, including computational neuroscience, genomics, and the study of protein structure and function. Building on my theoretical and applied work, I also look forward to extending this research to other scientific domains through interdisciplinary collaborations.

[4] Available at github.com/vincisLab/neuron-response-classification

[5] The dataset is available at https://doi.org/10.6080/K0B27SHN.

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